# 6: SIMULTANEOUS RANDOM VARIABLES

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Litterature:

H. H. Hamedani, *Probability and Statistics: The Science of Uncertainty*, 3rd ed., https://www.probabilitycourse.com

Course: Stokastisk modellering og behandling (SWSTS-01)

## Joint Probability Mass Function (PMF)

# Definition

For **discrete random variables** $X$ and $Y$, the **joint probability mass function (PMF)** is a function

$$f_{XY} : \mathbb{R}^2 \to \mathbb{R}$$

defined as

$$f_{XY}(x,y) = \begin{cases} P(X = x, Y = y), & (x,y) \in R_{XY} \\ 0, & \text{otherwise} \end{cases}$$

where

$$R_{XY} = \{(x,y) \mid P(X = x, Y = y) > 0\}$$

is the **joint support**.

# Probability of a Region

For any set $A \subset \mathbb{R}^2$

$$P((X,Y) \in A) = \sum_{(x_i,y_j) \in A \cap R_{XY}} P_{XY}(x_i, y_j).$$

## Joint Probability Mass Function (PMF)

# Properties of a Valid Joint PMF

### 1. Non-negativity

$$f_{XY}(x, y) \geq 0 \quad \text{for all } (x, y) \in \mathbb{R}^2$$

### 2. Normalization (Total Probability is 1)

$$\sum_x \sum_y f_{XY}(x, y) = 1$$

(or equivalently)

$$\sum_{(x,y) \in R_{XY}} f_{XY}(x, y) = 1$$

## Joint Probability Mass Function (PMF)

### Marginal PMFs

A Joint PMF contains the full information about the pair of random variables X and Y. From this we can extract the PMFs of X or Y separatly by use of marginal PMFs.

**Using Function Notation:**

$$f_X(x) = \sum_y f_{X,Y}(x,y), \quad f_Y(y) = \sum_x f_{X,Y}(x,y)$$

**Using Probability Notation with Explicit Ranges:**

$$P_X(x) = \sum_{y_j \in R_Y} P_{XY}(x, y_j), \quad \text{for any } x \in R_X$$

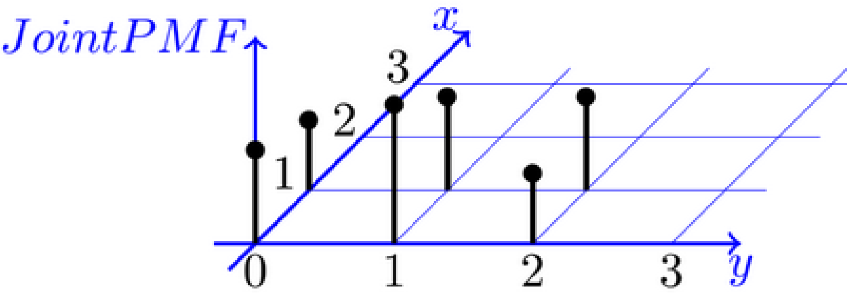
$$P_Y(y) = \sum_{x_i \in R_X} P_{XY}(x_i, y), \quad \text{for any } y \in R_Y$$

## Joint Probability Mass Function (PMF)

**Example:** A machine is monitored using two sensors

X: **Temperature Sensor** and Y: **Vibration Sensor**

| | $Y = 0$ (Normal) | $Y = 1$ (Slight anomaly) | $Y = 3$ (Severe anomaly) |
|---|---|---|---|
| $X = 0$ (Normal) | $1/6$ | $1/4$ | $1/8$ |
| $X = 1$ (Overheat) | $1/8$ | $1/6$ | $1/6$ |




**a.** What is the probability that the temperature is normal and vibration is not severe?
(i.e., $P(X = 0, Y \leq 1)$)

**b.** Find the **marginal PMFs** of $X$ and $Y$.

**c.** What is the probability that the vibration sensor shows a **slight anomaly**, given that the machine is **not overheating**?
(i.e., $P(Y = 1 \mid X = 0)$)

**d.** Are the temperature and vibration sensor readings **statistically independent**?

## Joint Probability Mass Function (PMF)

### a. Compute $P(X = 0, Y \leq 1)$

$$P(X = 0, Y \leq 1) = P_{XY}(0,0) + P_{XY}(0,1) = \frac{1}{6} + \frac{1}{4} = \frac{5}{12} \approx 0.417$$

### b. Marginal PMFs

- Temperature Sensor (X):

$$P_X(0) = \frac{1}{6} + \frac{1}{4} + \frac{1}{8} = \frac{13}{24} \approx 54.2\% \rightarrow \text{Temperature is normal}$$

$$P_X(1) = \frac{1}{8} + \frac{1}{6} + \frac{1}{6} = \frac{11}{24} \approx 45.8\% \rightarrow \text{Temperature is overheating}$$

- Vibration Sensor (Y):

$$P_Y(0) = \frac{1}{6} + \frac{1}{8} = \frac{7}{24}$$

$$P_Y(1) = \frac{1}{4} + \frac{1}{6} = \frac{5}{12}$$

$$P_Y(2) = \frac{1}{8} + \frac{1}{6} = \frac{7}{24}$$

### c. Compute $P(Y = 1 \mid X = 0)$

$$P(Y = 1 \mid X = 0) = \frac{P_{XY}(0,1)}{P_X(0)} = \frac{\frac{1}{4}}{\frac{13}{24}} = \frac{6}{13} \approx 0.462 \text{ or } 46.2\%$$

### d. Are X and Y independent?

Check whether:

$$P(Y = 1 \mid X = 0) = P_Y(1)$$

$$P(Y = 1 \mid X = 0) = \frac{6}{13}, \quad P_Y(1) = \frac{5}{12}$$

Since these are **not equal**, $X$ and $Y$ are **not independent**.

# Quiz

Two joint random variables $X$ and $Y$ have the following joint PMF $f_{X,Y}(x,y)$:

| $X\backslash Y$ | **1** | **2** | **3** |
|---|---|---|---|
| 0 | 1/10 | 2/10 | 0/10 |
| 1 | 3/10 | 1/10 | 3/10 |

**What is $f_{X|Y}(1 \mid 2)$?**

*(This is the* ***conditional probability*** *that $X = 1$ given that $Y = 2$)*

A: $1/10$

B: $3/10$

C: $1/3$

## Joint Cumulative Distributive Function (CDF)

For two random variables $X$ and $Y$, the **joint cumulative distribution function** $F_{X,Y}(x, y)$ is defined as:

$$F_{X,Y}(x, y) = P(X \leq x, Y \leq y)$$

### Sum of PMF Values:

For discrete random variables, the joint CDF is computed by summing the joint PMF values over all values $x_i \leq x$, $y_i \leq y$:

$$F_{X,Y}(x, y) = \sum_{x_i \leq x} \sum_{y_i \leq y} f_{X,Y}(x_i, y_i)$$

### Key Properties:

- **Lower bounds:**

$$F_{X,Y}(-\infty, y) = 0, \quad F_{X,Y}(x, -\infty) = 0$$

- **Upper bound (total probability):**

$$F_{X,Y}(\infty, \infty) = 1$$

## Joint Cumulative Distributive Function (CDF)

**Example:** A machine is monitored using two sensors

X: **Temperature Sensor** and Y: **Vibration Sensor**

| | $Y = 0$ (Normal) | $Y = 1$ (Slight anomaly) | $Y = 3$ (Severe anomaly) |
|---|---|---|---|
| $X = 0$ (Normal) | $1/6$ | $1/4$ | $1/8$ |
| $X = 1$ (Overheat) | $1/8$ | $1/6$ | $1/6$ |

**Compute** $F_{X,Y}(1,1)$

Definition

$$F_{X,Y}(x,y) = \sum_{x_i \le x} \sum_{y_i \le y} f_{XY}(x_i, y_i)$$

$$x = 1, \quad y = 1$$

so we include

$$(0,0), (1,0), (0,1), (1,1)$$

Thus

$$F_{X,Y}(1,1) = \frac{1}{6} + \frac{1}{8} + \frac{1}{4} + \frac{1}{6} \quad = \frac{17}{24}$$

## Joint Cumulative Distributive Function (CDF)

### Joint CDF Calculation

$F_{X,Y}(0,0)$

$$= \frac{1}{6}$$

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$F_{X,Y}(0,1)$

$$= \frac{1}{6} + \frac{1}{4} = \frac{5}{12}$$

---

$F_{X,Y}(0,3)$

$$= \frac{1}{6} + \frac{1}{4} + \frac{1}{8} = \frac{13}{24}$$

---

$F_{X,Y}(1,0)$

$$= \frac{1}{6} + \frac{1}{8} = \frac{7}{24}$$

---

$F_{X,Y}(1,1)$

$$= \frac{1}{6} + \frac{1}{8} + \frac{1}{4} + \frac{1}{6} = \frac{17}{24}$$

---

$F_{X,Y}(1,3)$

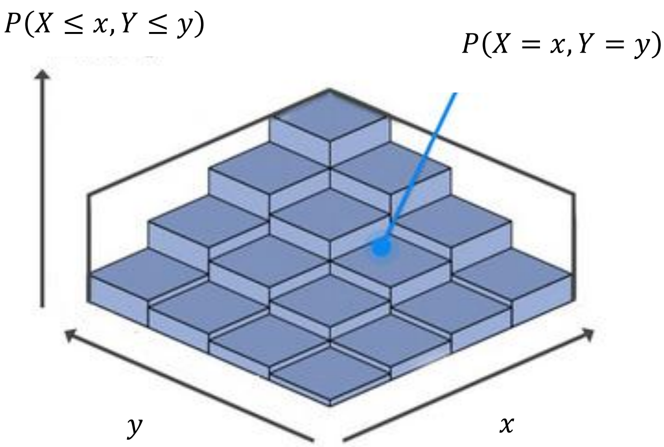
$$= 1$$

## Joint Cumulative Distributive Function (CDF)

When we plot the joint CDF surface

$$z = F_{XY}(x, y),$$

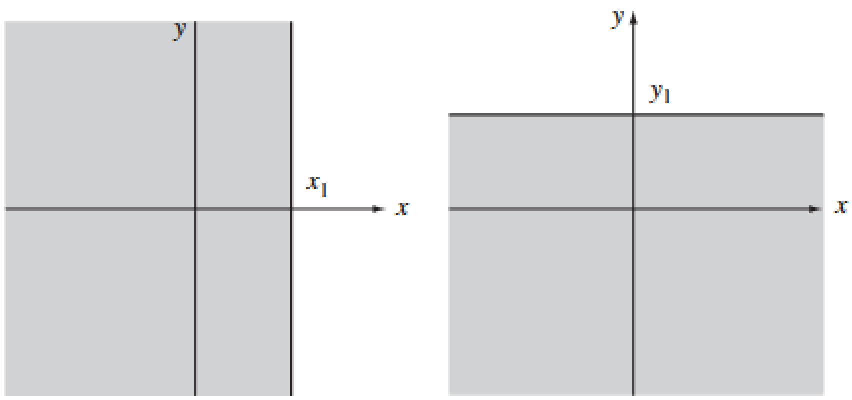
the **height is defined to be that probability**.




and

$$F_{XY}(x, y) = P((X, Y) \in (-\infty, x] \times (-\infty, y]).$$

The **marginal** cdf's are the probabilities of these half-planes:




$F_X(x_1) = P[X \leqslant x_1, Y < \infty]$ $\quad$ $F_Y(y_1) = P[X < \infty, Y \leqslant y_1]$

## Independence (Discrete Case)

### Independence via PMF or CDF

Two random variables $X$ and $Y$ are **independent** if:

$$P_{XY}(x,y) = P_X(x) \cdot P_Y(y) \quad \text{for all } x, y$$

Or equivalently:

$$F_{XY}(x,y) = F_X(x) \cdot F_Y(y) \quad \text{for all } x, y$$

### Conditional PMF and Independence

$$P(X = x_i \mid Y = y_j) = \frac{P(X = x_i \text{ and } Y = y_j)}{P(Y = y_j)} = \frac{P_{XY}(x_i, y_j)}{P_Y(y_j)}$$

If $X$ and $Y$ are independent, then:

$$P_{XY}(x_i, y_j) = P_X(x_i) \cdot P_Y(y_j)$$

Simplify

$$P(X = x_i \mid Y = y_j) = \frac{P_X(x_i) \cdot P_Y(y_j)}{P_Y(y_j)} = P_X(x_i)$$

## Independence (Discrete Case)

### Conditional Expectation under Independence

Using the conditional PMF:

$$E[X \mid Y = y] = \sum_{x \in \mathcal{R}_X} x \cdot P_{X|Y}(x \mid y) = \sum_{x \in \mathcal{R}_X} x \cdot P_X(x) = E[X]$$

Therefore, a function that does not change with Y:

$$E[X \mid Y] = E[X] \quad \text{(constant function of } Y\text{)}$$

### Product of Random Variables:

Using LOTUS (Law of the Unconscious Statistician) and independence:

$$\begin{aligned} E[XY] &= \sum_{x \in \mathcal{R}_X} \sum_{y \in \mathcal{R}_Y} xy \cdot P_{XY}(x, y) \\ &= \sum_x \sum_y xy \cdot P_X(x) P_Y(y) \\ &= \left( \sum_x x P_X(x) \right) \cdot \left( \sum_y y P_Y(y) \right) \\ &= E[X] \cdot E[Y] \end{aligned}$$

⚠️ If $E[XY] = E[X]E[Y]$, it **does not imply** that $X$ and $Y$ are independent.

## Dependence (Discrete Case)

### Example

Let:

$X \in \{-1, 0, +1\}, \quad P(X=-1) = P(X=+1) = \frac{1}{4},\ P(X=0) = \frac{1}{2}$

Define:

$$Y = X^2$$

Y distribution becomes

$$P(Y=0) = \tfrac{1}{2}, \quad P(Y=1) = \tfrac{1}{2}$$

**Expectations:**

$$E[X] = (-1) \cdot \tfrac{1}{4} + 0 \cdot \tfrac{1}{2} + 1 \cdot \tfrac{1}{4} = 0$$

$$E[Y] = (0^2) \cdot \tfrac{1}{2} + (1^2) \cdot \tfrac{1}{2} = \tfrac{1}{2}$$

$$E[XY] = E[X \cdot X^2] = E[X^3] = (-1)^3 \cdot \tfrac{1}{4} + 0 + (1)^3 \cdot \tfrac{1}{4} = -\tfrac{1}{4} + \tfrac{1}{4} = 0$$

$$E[X] \cdot E[Y] = 0 \cdot \tfrac{1}{2} = 0$$

**Conclusion:**

$$E[XY] = E[X]E[Y] \quad \text{but} \quad X \not\perp\!\!\!\perp Y$$

(Because, for example, $P(X=0) = \frac{1}{2}$ but $P(X=0 \mid Y=1) = 0$.)

## Quiz

### There might be an error in the following reasoning. Can you find it?

Let:

$$X \in \{-1, +1\}, \quad P(X = -1) = P(X = +1) = \tfrac{1}{2}$$

Define:

$$Y = X^2$$

**Distributions:**

$$Y = 1 \quad \text{with probability } 1 \quad \Rightarrow \quad P(Y = 1) = 1$$

**Expectations:**

$$\begin{aligned} \mathbb{E}[X] &= (-1) \cdot \tfrac{1}{2} + 1 \cdot \tfrac{1}{2} = 0 \\ \mathbb{E}[Y] &= 1 \\ \mathbb{E}[XY] &= \mathbb{E}[X \cdot X^2] = \mathbb{E}[X^3] = \mathbb{E}[X] = 0 \\ \mathbb{E}[X] \cdot \mathbb{E}[Y] &= 0 \cdot 1 = 0 \end{aligned}$$

**Conclusion:**

$$\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y] \quad \text{but} \quad X \not\perp Y$$

**A.** The expectations $\mathbb{E}[X]$, $\mathbb{E}[Y]$, and $\mathbb{E}[XY]$ were calculated incorrectly.

**B.** Since $Y = X^2$, it is a constant ($Y = 1$) random variable. Constant random variables are independent of everything, so the conclusion $X \not\perp Y$ is false.

**C.** Independence only requires $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$, so the conclusion was correct and there is no error.

# Dependence (Discrete Case)

## Dependence: What Changes?

- **Joint PMF**:

$$P_{XY}(x,y) \neq P_X(x) \cdot P_Y(y)$$

  The behavior of one variable **depends on** the value of the other.

- **Conditional PMF**:

$$P(X = x \mid Y = y) \neq P_X(x)$$

  The distribution of $X$ **changes with** $Y$.

- **Conditional Expectation**:

$$\mathbb{E}[X \mid Y = y] \neq \mathbb{E}[X]$$

  Now $\mathbb{E}[X \mid Y]$ is a **non-constant function of** $Y$.

## Implication on Expectation of Product

$$E[XY] \neq E[X]E[Y] \quad \Longrightarrow \quad X \text{ and } Y \text{ are dependent.}$$

# Functions of Two Random Variables

Let $X$ and $Y$ be two discrete random variables with joint PMF

$$P_{XY}(x,y) = P(X = x, Y = y).$$

Suppose we define a new random variable

$$Z = g(X,Y),$$

where

$$g : \mathbb{R}^2 \to \mathbb{R}.$$

## PMF of $Z$

Define the set

$$A_z = \{(x_i, y_j) \in R_{XY} : g(x_i, y_j) = z\}.$$

Then the PMF of $Z$ is obtained by summing the joint probabilities over all such pairs:

$$P_Z(z) = \sum_{(x_i, y_j) \in A_z} P_{XY}(x_i, y_j).$$

## Functions of Two Random Variables

### Example

Let

$$X, Y \in \{0, 1\}$$

with joint PMF

| | $Y = 0$ | $Y = 1$ |
|---|---|---|
| $X = 0$ | $1/4$ | $1/4$ |
| $X = 1$ | $1/4$ | $1/4$ |

Thus

$$p_{XY}(x, y) = \frac{1}{4}, \quad x, y \in \{0, 1\}.$$

Define $Z = X + Y$ with range $Z \in \{0,1,2\}$. Then PMF of Z is:

$$p_Z(z) = \sum_{x+y=z} p_{XY}(x, y).$$

$$P_Z(0) = P_{XY}(0,0) = \frac{1}{4}$$

$$P_Z(1) = P_{XY}(0,1) + P_{XY}(1,0) = \frac{1}{2}$$

$$P_Z(2) = P_{XY}(1,1) = \frac{1}{4}$$

This is a **discrete convolution** of the PMFs of $X$ and $Y$.

# Functions of Two Random Variables

## Convolution in general when X and Y are independent Variables

- We're interested in all the **pairs** $(x, y)$ such that $x + y = z$.
- Since $Z = X + Y$, for each possible value of $x$, we want the probability that:
  - $X = x$, and
  - $Y = z - x$, because then $X + Y = x + (z - x) = z$.

Because $X$ and $Y$ are **independent**, we can write:

$$P(X = x \text{ and } Y = z - x) = P(X = x) \cdot P(Y = z - x)$$

We then **sum over all possible values of** $x$ to get all ways of achieving the total sum $z$.

The full formula for $P(Z = z)$ becomes:

$$P(Z = z) = \sum_{x} P(X = x) \cdot P(Y = z - x)$$

# Functions of Two Random Variables

**Many real phenomena are additive:**

### Measurement noise

$$\text{measurement} = \text{true value} + \text{noise}$$

### Total cost

$$\text{total cost} = \text{materials} + \text{labor}$$

### Queueing time

$$\text{total waiting time} = W_1 + W_2 + W_3$$

### Random walks

$$S_n = X_1 + X_2 + \cdots + X_n$$

Because **many systems accumulate effects**, addition naturally appears.

# Functions of Two Random Variables

## Linearity of Expectation (additivity)

Let $X$ and $Y$ be discrete random variables with joint PMF

$$p_{X,Y}(x,y) = P(X = x, Y = y).$$

Then

$$E[X+Y] = E[X] + E[Y].$$

Proof (use LOTUS)

$$E[X+Y] = \sum_{x \in R_X} \sum_{y \in R_Y} (x+y)\, p_{X,Y}(x,y).$$

Using distributivity,

$$(x+y)p_{X,Y}(x,y) = x\, p_{X,Y}(x,y) + y\, p_{X,Y}(x,y).$$

We get

$$E[X+Y] = \sum_x \sum_y x\, p_{X,Y}(x,y) + \sum_x \sum_y y\, p_{X,Y}(x,y).$$

Rearrange the sums

$$E[X+Y] = \sum_x x \left( \sum_y p_{X,Y}(x,y) \right) + \sum_y y \left( \sum_x p_{X,Y}(x,y) \right).$$

Recognize the marginal PMFs

$$E[X+Y] = \sum_x x\, p_X(x) + \sum_y y\, p_Y(y).$$

Therefore

$$E[X+Y] = E[X] + E[Y].$$

## Joint Probability Density Function (PDF)

### Definition

For continuous random variables $X$ and $Y$, the **joint probability density function (PDF)** is a function

$$f_{XY} : \mathbb{R}^2 \to \mathbb{R}$$

defined as:

$$f_{XY}(x, y) = \begin{cases} \text{density value}, & \text{if } (x, y) \in R_{XY} \\ 0, & \text{otherwise} \end{cases}$$

The probability that $(X, Y)$ lies within a region $A \subseteq \mathbb{R}^2$ is given by:

$$P((X, Y) \in A) = \iint_A f_{XY}(x, y)\, dx\, dy$$

### Properties of a Valid Joint PDF

1. **Non-negativity:**

$$f_{XY}(x, y) \geq 0 \quad \text{for all } (x, y) \in \mathbb{R}^2$$

2. **Normalization (Total Probability is 1):**

$$\iint_{\mathbb{R}^2} f_{XY}(x, y)\, dx\, dy = 1$$

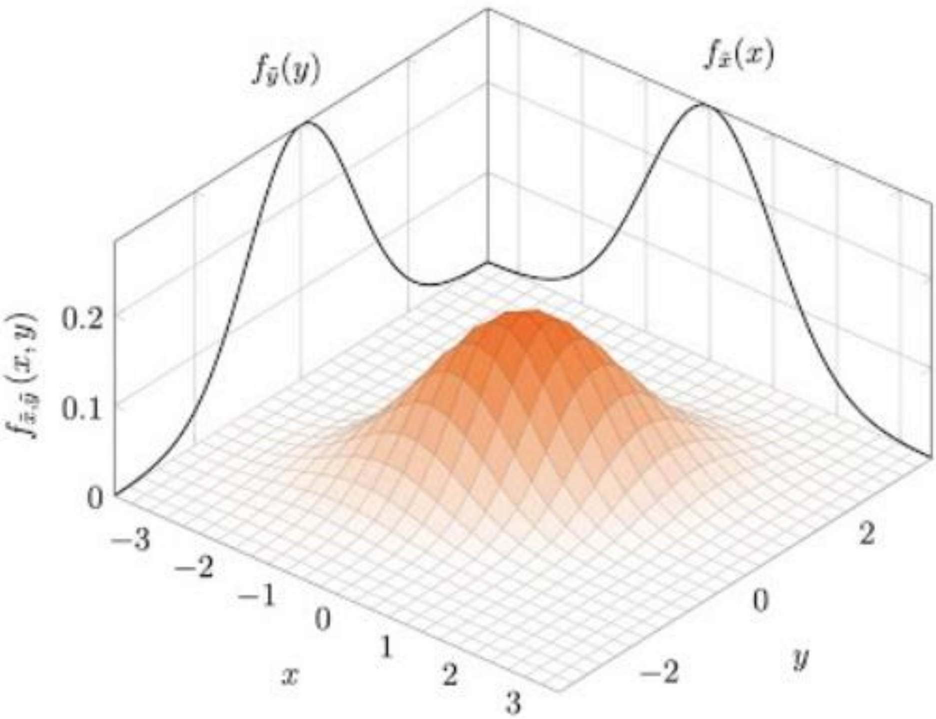
## Joint Probability Density Function (PDF)

### Marginal PDFs

Similar to marginal PMFs with sums replaced by integrals:

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)\,dy, \quad \text{for all } x$$

$$f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)\,dx, \quad \text{for all } y$$



## Joint Probability Density Function (PDF)

**Example:** Joint PDF and Marginals

Let $X$ and $Y$ be jointly continuous random variables with joint probability density function:

$$f_{XY}(x,y) = \begin{cases} x + cy^2, & 0 \le x \le 1,\ 0 \le y \le 1, \\ 0, & \text{otherwise.} \end{cases}$$

### (a) Find the constant $c$

$$\iint_{\mathbb{R}^2} f_{XY}(x,y)\,dx\,dy = 1. \quad \Rightarrow$$

$$\int_0^1 \int_0^1 (x + cy^2)\,dx\,dy = \int_0^1 \left[\frac{1}{2} + cy^2\right] dy = \left[\frac{1}{2}y + \frac{c}{3}y^3\right]_0^1 = \frac{1}{2} + \frac{c}{3}.$$

Set this equal to 1: $\frac{1}{2} + \frac{c}{3} = 1 \Rightarrow c = \frac{3}{2}.$

## Joint Probability Density Function (PDF)

### (b) Find the probability:

$$P(0 \le X \le \tfrac{1}{2},\, 0 \le Y \le \tfrac{1}{2}) = \int_0^{1/2} \int_0^{1/2} \left(x + \frac{3}{2}y^2\right) dx\, dy.$$

$$= \int_0^{1/2} \left[\frac{1}{8} + \frac{3}{4}y^2\right] dy = \left[\frac{1}{8}y + \frac{1}{4}y^3\right]_0^{1/2} = \frac{1}{16} + \frac{1}{32} = \frac{3}{32}.$$

### (c) Find the marginal PDF $f_X(x)$

$$f_X(x) = \int_{-\infty}^{\infty} f_{XY}(x,y)\, dy = \int_0^1 \left(x + \frac{3}{2}y^2\right) dy.$$

$$= \left[xy + \frac{1}{2}y^3\right]_0^1 = x + \frac{1}{2}, \quad \text{for } 0 \le x \le 1.$$

So,

$$f_X(x) = \begin{cases} x + \frac{1}{2}, & 0 \le x \le 1, \\ 0, & \text{otherwise.} \end{cases}$$

## Joint Cumulative Distributive Function (CDF)

For two continuous random variables $X$ and $Y$, the **joint cumulative distribution function (CDF)** $F_{X,Y}(x,y)$ is defined as:

$$F_{X,Y}(x,y) = P(X \leq x, Y \leq y)$$

### Integral of PDF Values:

For continuous random variables, the joint CDF is computed by integrating the joint PDF over all values $u \leq x$, $v \leq y$:

$$F_{X,Y}(x,y) = \int_{-\infty}^{y} \int_{-\infty}^{x} f_{X,Y}(u,v)\, du\, dv$$

### Key Properties:

- **Lower bounds:**

$$F_{X,Y}(-\infty, y) = 0, \quad F_{X,Y}(x, -\infty) = 0$$

- **Upper bound (total probability):**

$$F_{X,Y}(\infty, \infty) = 1$$

### Relationship to Joint PDF:

$$f_{X,Y}(x,y) = \frac{\partial^2}{\partial x\, \partial y} F_{X,Y}(x,y)$$

## Independence (Continuous Case)

### Independence via PDF or CDF

Two random variables $X$ and $Y$ are **independent** if:

$$f_{XY}(x, y) = f_X(x) \cdot f_Y(y) \quad \text{for all } x, y$$

Or equivalently:

$$F_{XY}(x, y) = F_X(x) \cdot F_Y(y) \quad \text{for all } x, y$$

### Conditional PDF and Independence

$$f_{X|Y}(x \mid y) = \frac{f_{XY}(x, y)}{f_Y(y)}$$

If $X$ and $Y$ are independent, then:

$$f_{X|Y}(x \mid y) = \frac{f_X(x) \cdot f_Y(y)}{f_Y(y)} = f_X(x)$$

## Independence (Continuous Case)

### Conditional Expectation under Independence

Using the conditional PDF:

$$E[X \mid Y = y] = \int_{-\infty}^{\infty} x \cdot f_{X|Y}(x \mid y)\, dx$$

$$= \int_{-\infty}^{\infty} x \cdot f_X(x)\, dx = E[X]$$

Therefore, if $X$ and $Y$ are independent:

$$E[X \mid Y] = E[X] \quad (\text{as a constant function of } Y)$$

### Product of Random Variables

Using LOTUS (Law of the Unconscious Statistician) and independence:

$$E[XY] = \iint xy \cdot f_{XY}(x, y)\, dxdy = \iint xy \cdot f_X(x) f_Y(y)\, dxdy$$

$$= \left(\int x f_X(x) dx\right) \cdot \left(\int y f_Y(y) dy\right) = E[X] \cdot E[Y]$$

⚠️ If $E[XY] = E[X]E[Y]$, it **does not imply** that $X$ and $Y$ are independent.

# Functions of Two Random Variables

## The Method of Transformations

Suppose $X, Y$ are continuous random variables with joint CDF

$$F_{XY}(x, y) = P(X \leq x,\ Y \leq y)$$

and joint PDF

$$f_{XY}(x, y).$$

We define new random variables

$$Z = g_1(X, Y), \qquad W = g_2(X, Y).$$

Equivalently,

$$(Z, W) = g(X, Y)$$

where

$$g : \mathbb{R}^2 \to \mathbb{R}^2.$$

Let

$$h = g^{-1}.$$

Then

$$(X, Y) = h(Z, W)$$

with

$$x = h_1(z, w), \qquad y = h_2(z, w).$$

# Functions of Two Random Variables

## Transform the CDF

$$F_{ZW}(z,w) = P(Z \le z,\ W \le w)$$

$$= P(g_1(X,Y) \le z,\ g_2(X,Y) \le w)$$

$$= P\big(X \le h_1(z,w),\ Y \le h_2(z,w)\big)$$

$$= F_{XY}\big(h_1(z,w),\ h_2(z,w)\big),$$

where $h = g^{-1}$.

## Differentiate the CDF

$$f_{ZW}(z,w) = \frac{\partial^2}{\partial z\, \partial w} F_{XY}(h_1(z,w), h_2(z,w)).$$

## Apply the Chain Rule

First differentiate with respect to $z$:

$$\frac{\partial}{\partial z} F_{XY}(h_1(z,w), h_2(z,w))$$

Using the multivariable chain rule:

$$= \frac{\partial F_{XY}}{\partial x}\frac{\partial h_1}{\partial z} + \frac{\partial F_{XY}}{\partial y}\frac{\partial h_2}{\partial z}.$$

Now differentiate with respect to $w$.

After applying the chain rule again and simplifying, we obtain

$$f_{ZW}(z,w) = f_{XY}(h_1(z,w), h_2(z,w)) \left( \frac{\partial h_1}{\partial z}\frac{\partial h_2}{\partial w} - \frac{\partial h_1}{\partial w}\frac{\partial h_2}{\partial z} \right).$$

## Replace with the Jacobian:

$$\frac{\partial h_1}{\partial z}\frac{\partial h_2}{\partial w} - \frac{\partial h_1}{\partial w}\frac{\partial h_2}{\partial z} = \det \begin{bmatrix} \frac{\partial h_1}{\partial z} & \frac{\partial h_1}{\partial w} \\ \frac{\partial h_2}{\partial z} & \frac{\partial h_2}{\partial w} \end{bmatrix}.$$

# Functions of Two Random Variables

## Example: Distribution of the Sum

Let $X$ and $Y$ be continuous random variables with joint PDF $f_{XY}(x, y)$.

Define

$$Z = X + Y.$$

To apply the **transformation method**, introduce a second variable

$$W = X.$$

Thus the transformation

$$g(X, Y) = (Z, W)$$

is

$$\begin{cases} z = x + y \\ w = x \end{cases}$$

## Inverse Transformation

Solve for $x, y$:

$$\begin{cases} x = w \\ y = z - w \end{cases}$$

Thus

$$(X, Y) = h(Z, W)$$

with

$$h_1(z, w) = w, \qquad h_2(z, w) = z - w.$$

# Functions of Two Random Variables

## Jacobian

$$J = \det \begin{bmatrix} \frac{\partial h_1}{\partial z} & \frac{\partial h_1}{\partial w} \\ \frac{\partial h_2}{\partial z} & \frac{\partial h_2}{\partial w} \end{bmatrix} = \det \begin{bmatrix} 0 & 1 \\ 1 & -1 \end{bmatrix} = -1$$

so

$$|J| = 1.$$

## Joint PDF of $Z, W$

Using the transformation formula

$$f_{ZW}(z,w) = f_{XY}(h_1(z,w), h_2(z,w))|J|$$

gives

$$f_{ZW}(z,w) = f_{XY}(w, z-w).$$

## PDF of $Z$

To obtain the distribution of $Z$, integrate out $W$:

$$f_Z(z) = \int_{-\infty}^{\infty} f_{ZW}(z,w)\, dw = \int_{-\infty}^{\infty} f_{XY}(w, z-w)\, dw.$$

# Functions of Two Random Variables

## Independent Case

If $X$ and $Y$ are independent,

$$f_{XY}(x, y) = f_X(x) f_Y(y).$$

Thus

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(w) f_Y(z - w) \, dw.$$

This integral is called the **convolution** of $f_X$ and $f_Y$:

$$f_Z = f_X * f_Y.$$

# Functions of Two Random Variables

## Define $X$ and $Y$ with constant marginals

$$f_X(x) = \begin{cases} 1, & 0 \le x \le 1 \\ 0, & \text{otherwise} \end{cases} \qquad f_Y(y) = \begin{cases} 1, & 0 \le y \le 1 \\ 0, & \text{otherwise.} \end{cases}$$

**Convolution:** $Z = X + Y$

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(w)\, f_Y(z-w)\, dw = \int_{-\infty}^{\infty} 1_{\{0 \le w \le 1\}}\, 1_{\{0 \le z-w \le 1\}}\, dw.$$

The integrand is 1 exactly when both conditions hold:

$$0 \le w \le 1 \quad \text{and} \quad 0 \le z - w \le 1 \iff z - 1 \le w \le z.$$

So the valid $w$-range is the intersection

$$w \in [0,1] \cap [z-1,\, z].$$

Therefore $f_Z(z)$ equals the **length** of this intersection.

**Case 1:** $0 \le z \le 1$

$$[0,1] \cap [z-1, z] = [0, z] \Rightarrow f_Z(z) = z.$$

**Case 2:** $1 \le z \le 2$

$$[0,1] \cap [z-1, z] = [z-1, 1] \Rightarrow f_Z(z) = 2 - z.$$

**Otherwise**

No overlap, so $f_Z(z) = 0$.

## Final result (triangle PDF)

$$f_Z(z) = \begin{cases} z, & 0 \le z \le 1, \\ 2 - z, & 1 \le z \le 2, \\ 0, & \text{otherwise.} \end{cases}$$

## Covariance

**Definition:**

For two random variables $X$ and $Y$, the **covariance** is defined as:

$$\mathrm{Cov}(X,Y) = \mathbb{E}[(X-\mathbb{E}[X])(Y-\mathbb{E}[Y])] = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$$

**Interpretation:**

- Measures linear relationship between $X$ and $Y$.
- Positive covariance: large $X$ associated with large $Y$.
- Negative covariance: large $X$ associated with small $Y$.

### Properties of Covariance

Let $a, b \in \mathbb{R}$, and $X, Y, Z$ be random variables:

1. $\mathrm{Cov}(X,X) = \mathrm{Var}(X)$
2. If $X$ and $Y$ are independent, then $\mathrm{Cov}(X,Y) = 0$

**Discrete Case:** LOTUS

$$\mathrm{Cov}(X,Y) = \sum_x \sum_y (x-\mu_X)(y-\mu_Y) \cdot P(X=x, Y=y)$$

**Continuous Case:**

$$\mathrm{Cov}(X,Y) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} (x-\mu_X)(y-\mu_Y) \cdot f_{X,Y}(x,y)\,dx\,dy$$

# Covariance

### Variance of a Sum

Let $Z = X + Y$ be the sum of two random variables.

$$\mathrm{Var}(X+Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) + 2\mathrm{Cov}(X,Y)$$

### Variance of a Linear Combination

For $Z = aX + bY$, we have:

$$\mathrm{Var}(aX+bY) = a^2\mathrm{Var}(X) + b^2\mathrm{Var}(Y) + 2ab\,\mathrm{Cov}(X,Y)$$

### Uncorrelated vs Independent

- If $X$ and $Y$ are independent, then $\mathrm{Cov}(X,Y) = 0 \Rightarrow$ $X$ and $Y$ are uncorrelated.

⚠️ The converse is **not** necessarily true: uncorrelated does **not imply** independence.

### Implication:

$$\text{If } \mathrm{Cov}(X,Y) = 0, \text{ then } \mathrm{Var}(X+Y) = \mathrm{Var}(X) + \mathrm{Var}(Y)$$

## Correlation Coefficient

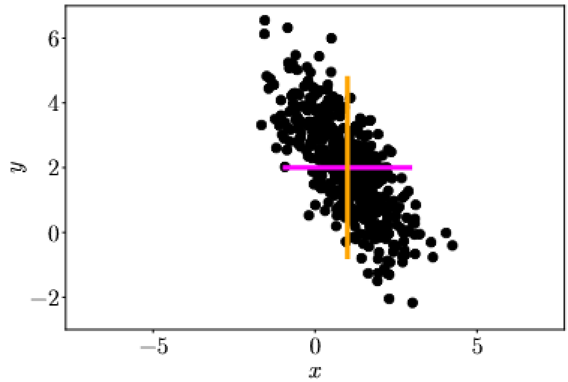
### Definition:

The **correlation coefficient** $\rho_{XY}$ is defined as:

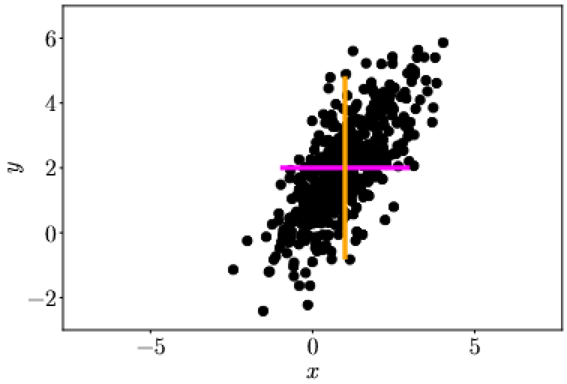
$$\rho(X,Y) = \frac{\text{Cov}(X,Y)}{\sqrt{\text{Var}(X)}\sqrt{\text{Var}(Y)}} = \frac{\text{Cov}(X,Y)}{\sigma_X \sigma_Y}$$

### Properties:

1. $-1 \leq \rho(X,Y) \leq 1$
2. $\rho(X,Y) = 1 \Rightarrow Y = aX + b, a > 0$
3. $\rho(X,Y) = -1 \Rightarrow Y = aX + b, a < 0$
4. $\rho(aX + b, cY + d) = \rho(X,Y)$ for $a, c > 0$



(a) $x$ and $y$ are negatively correlated.



(b) $x$ and $y$ are positively correlated.

## Correlation Coefficient

# Example: Python

```python
import numpy as np
import matplotlib.pyplot as plt

# --- Correlation for simple example arrays ---
X = np.array([1, 2, 3, 4])
Y = np.array([2, 4, 6, 8])

rho = np.corrcoef(X, Y)[0, 1]
```

## Correlation Coefficient

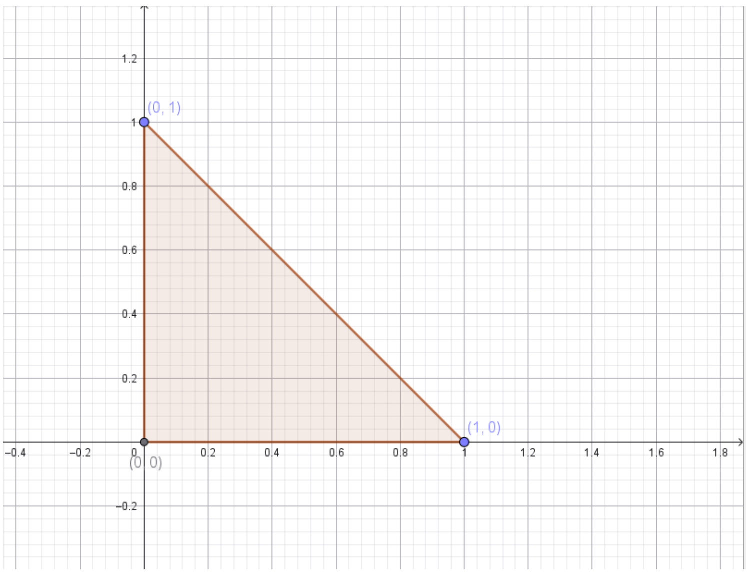
### Example: Joint PDF

$X, Y$ have joint pdf

$$f_{XY}(x,y) = \begin{cases} 2, & x > 0,\ y > 0,\ x + y \le 1, \\ 0, & \text{otherwise.} \end{cases}$$

Find $\mathrm{Cov}(X,Y)$ and $\rho(X,Y)$.

The support is the right triangle with vertices $(0,0)$, $(1,0)$, $(0,1)$.




The triangle's area is $1/2$. Since the density is constant $2$,

$$\iint f_{XY} = 2 \times \text{area} = 2 \cdot \tfrac{1}{2} = 1.$$

So it's a valid pdf.

# Correlation Coefficient

## Marginal pdf's

For a fixed $x \in [0,1]$, $y$ ranges from $0$ to $1-x$:

$$f_X(x) = \int_0^{1-x} 2\,dy = 2(1-x), \qquad 0 \le x \le 1.$$

By symmetry,

$$f_Y(y) = 2(1-y), \qquad 0 \le y \le 1.$$

## Means and variances

$$E[X] = \int_0^1 x\,f_X(x)\,dx = \int_0^1 x \cdot 2(1-x)\,dx = 2\int_0^1 (x - x^2)\,dx = 2\left[\tfrac{1}{2} - \tfrac{1}{3}\right] = \tfrac{1}{3}$$

By symmetry $E[Y] = \frac{1}{3}$.

$$E[X^2] = \int_0^1 x^2 f_X(x)\,dx = 2\int_0^1 (x^2 - x^3)\,dx = 2\left[\tfrac{1}{3} - \tfrac{1}{4}\right] = \tfrac{1}{6}.$$

Hence

$$\mathrm{Var}(X) = E[X^2] - E[X]^2 = \tfrac{1}{6} - \tfrac{1}{9} = \tfrac{1}{18}.$$

Again by symmetry, $\mathrm{Var}(Y) = \frac{1}{18}$.

## $E[XY]$

$$\begin{aligned} E[XY] &= \int_0^1 \int_0^{1-x} 2\,x\,y\,dy\,dx = \int_0^1 2x\left[\tfrac{y^2}{2}\right]_0^{1-x} dx \\ &= \int_0^1 x(1-x)^2\,dx = \int_0^1 (x - 2x^2 + x^3)\,dx \\ &= \left[\tfrac{x^2}{2} - \tfrac{2x^3}{3} + \tfrac{x^4}{4}\right]_0^1 = \tfrac{1}{2} - \tfrac{2}{3} + \tfrac{1}{4} = \tfrac{1}{12}. \end{aligned}$$

$$\mathrm{Cov}(X,Y) = E[XY] - E[X]E[Y] = \tfrac{1}{12} - \left(\tfrac{1}{3}\right)^2 = \tfrac{1}{12} - \tfrac{1}{9} = -\tfrac{1}{36}.$$

$$\rho(X,Y) = \frac{\mathrm{Cov}(X,Y)}{\sqrt{\mathrm{Var}(X)\,\mathrm{Var}(Y)}} = \frac{-\frac{1}{36}}{\sqrt{\frac{1}{18} \cdot \frac{1}{18}}} = \frac{-\frac{1}{36}}{\frac{1}{18}} = -\tfrac{1}{2}.$$

## Quiz

**Two discrete-time signals $x[n]$ and $y[n]$ have positive covariance. What does this imply about their behavior over time?**

A. When $x[n]$ is high, $y[n]$ tends to be low.

B. $x[n]$ and $y[n]$ tend to vary in the same direction.

C. There is no relation between $x[n]$ and $y[n]$.

D. $x[n]$ and $y[n]$ are independent.

## Quiz

**You work with two independent sensor readings $X$ and $Y$. If $Var(X) = 4$ and $Var(Y) = 9$, what is the variance of $X + Y$?**

A: 13

B: 5

C: 15

## Quiz

Let $X$ and $Y$ be random variables such that:

- $\mathrm{Var}(X) = 1$
- $\mathrm{Var}(Y) = 4$
- $\rho_{XY} = -0.75$

What is the **covariance** between $X$ and $Y$?

A. $-0.75$

B. $-1.5$

C. $-3.0$

D. Cannot be determined from the information given